

Solution Requirements

Filled project document for GitHub and SkillWallet submission

Date	28 June 2026	Team ID	To be updated in SkillWallet
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction	Team Members	Devireddy Prasanna, Chokka Abitha, Navya Borugadda, Bodi Varshitha, Mulakala Harika Venkata Sridevi

Functional Requirements

FR No.	Functional Requirement	Sub Requirement / Story
FR-1	Dataset management	Collect, load, inspect, and preserve the flood prediction dataset.
FR-2	Data preprocessing	Handle missing values, scale numerical columns, and split data.
FR-3	Model training	Train Decision Tree, Random Forest, KNN, and XGBoost classifiers.
FR-4	Model evaluation	Generate accuracy, confusion matrix, and classification report.
FR-5	Web prediction	Accept weather inputs and return flood/no-flood prediction.
FR-6	Repository submission	Upload code, dataset, report, results, and references to GitHub.

Non-functional Requirements

NFR No.	Requirement	Description
NFR-1	Usability	The UI must be simple for authorities and mentors to operate.
NFR-2	Reliability	The saved model and scaler must reproduce notebook predictions.
NFR-3	Performance	Prediction response should be near-instant for a single region.
NFR-4	Scalability	Architecture should support IBM Cloud deployment.
NFR-5	Maintainability	Code should be readable, modular, and documented.
NFR-6	Security	Input validation should prevent invalid values and runtime errors.